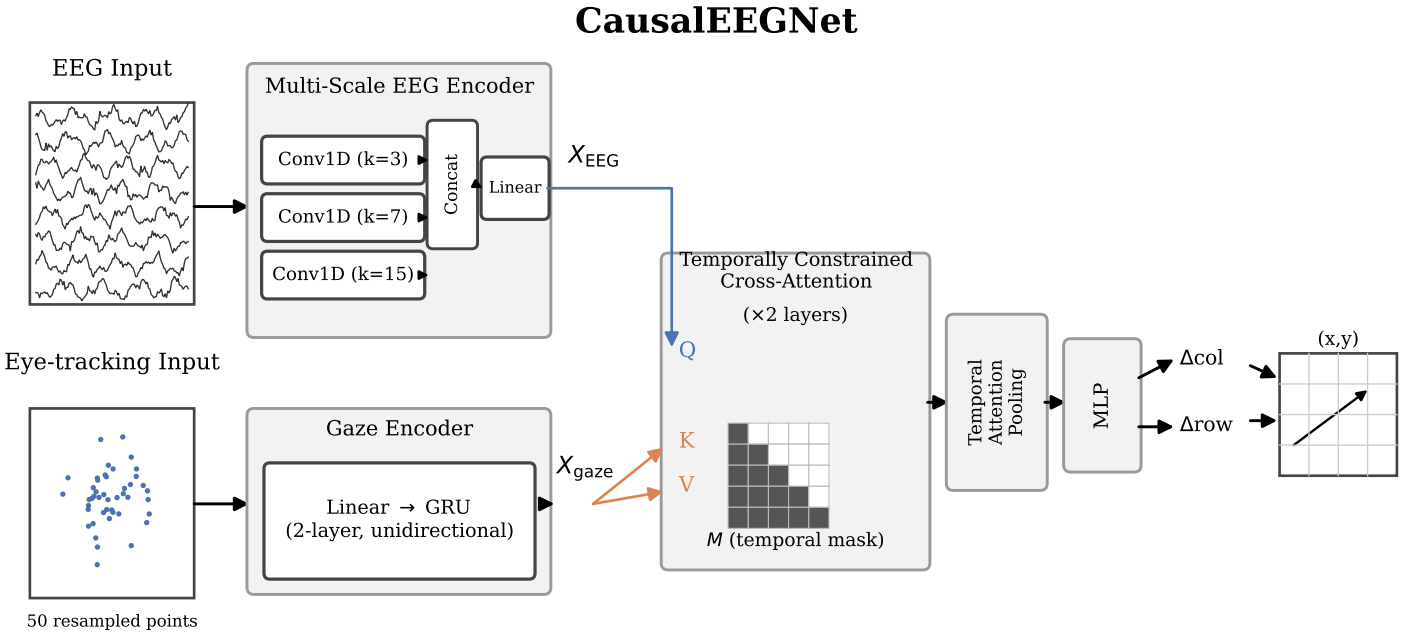
CausalEEGNet
EEG Input
Multi-Scale EEG Encoder
Conv1D (k=3)
Conv1D (k=7)
Conv1D (k=15)
Concat
Linear
X_{EEG}
Eye-tracking Input
Gaze Encoder
Linear → GRU
(2-layer, unidirectional)
X_{gaze}
50 resampled points
Temporally Constrained
Cross-Attention
(×2 layers)
Q
K
V
M (temporal mask)
Temporal
Attention
Pooling
MLP
Δcol
Δrow
(x,y)